MILITARY SPECIFICATION

RECEIVER, RADAR AUGMENTOR R/1623(*)/APN
MODULATOR, X-BAND BEACON MD-817(*)/APN
AND
ANTENNA, KA-BAND RADAR AS-2406(*)/APR

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall receive and process Ka-Band interrogations from the AN/SPN-42 Radar, part of the All-weather Carrier Landing System (ACLS).

1.2 Classification - The equipment covered by this specification shall consist of the following items:

Type Designation	Applicable Paragraph
Receiver Radar Augmentor R-1623(*)/APN	3.5.1
Modulator, X-Band Beacon MD-817(*)/APN	3.5.2
Antenna, Ka-Band Radar AS-2406(*)/APR	3.5.3

1.3 Associated Equipment - The augmentor receiver specified herein shall operate with the associated equipment listed in 6.7.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issues in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

Specifications

Military

MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment

FSC 5841

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Formal for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-B-23431	Beacon, Radar AN/APN-154 (V)

Naval Air Systems Command

AR-5	Microelectronic Devices used in Avionics Equipment, Procedures for Selection and Approval of
AR-8	Versatile Avionics Shop Test System, Avionics Systems Compatibility, General Requirements for
AR-9	Versatile Avionics Shop Test System Test Program, General Requirements for
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for

Standards

Military

MIL-STD-461	Electromagnetic Interference Characteristics Requirement for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definitions and Systems of Units, Electromagnetic Interference Technology
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of

2.1.1 Availability of Documents - (1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

1. Microelectronic Technology shall be considered and microelectronic items shall conform to requirements specified herein.

2. Other parts and materials requirements shall conform to Specification MIL-E-5400.

3. Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (See 6.5).

4. When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ non-repairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in Specification MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.2.2 Microelectronic Modular Assemblies - When used, Microelectronic Modular Assemblies shall meet the requirements of Specification AR-5.

3.2.3 Modules - The electronic portions of the equipment shall be functionally modularized in accordance with Specification AR-10.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed six pounds.

Weight of the individual items shall not exceed the values listed below:

Receiver, Augmentor	5.0 lbs

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

Modulator, X-Band	0.5 lb
Antenna, Ka-Band	0.5 lb

3.3.2 Reliability

3.3.2.1 Operating Life - The equipment shall have a minimum total operating life of 6,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.3.2.2 Reliability in Mean Time Between Failures (MTBF) - The specified mean (operating) time between failures shall be 1,000 hours when tested and accepted as outlined under the requirements of 4.4.2.

3.3.2.3 Time Totalizing Meter - The receiver shall contain a time totalizing meter in accordance with Specification MIL-M-7793.

3.3.2.4 Fault Indicator - A test signal, initiated by the push button switch/indicator lamp on the associated control panel, shall cause the receiver to generate triggers to the X-band beacon. The X-band replies detected in the X-band Modulator cause a "GO" indication at the control panel test lamp. The test lamp shall not light if the augmentor fails to respond to the test signal. In addition, a malfunction in the augmentor receiver shall be indicated by a latching type fault indicator.

3.3.3 Cabling and Connections

3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements of Specification MIL-E-5400.

3.3.5 Interference Control- The generation of radio interference by the augmentor receiver and the vulnerability of the receiver to radio interference shall meet the requirements of MIL-STD-461, MIL-STD-462, and MIL-STD-463.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.3.6 Provisions for Maintainability - The Maintainability Program built-in test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in Specification AR-10.

3.3.6.1 Compatibility with VAST - The equipment shall be compatible with the Versatile Avionic Shop Test System (VAST) and shall meet the requirements of Specification AR-8. When required by contract, VAST Test Programs shall be furnished in accordance with Specification AR-9. If VAST Test Programs exist for the equipment, and changes to the equipment are made which will affect the fault diagnosis procedure, changes to the existing Test Program shall be prepared as part of the equipment changes in accordance with Specification AR-9.

3.3.7 Nomenclature and Nameplates - Nomenclature assignments and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.3.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (25°C ±5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	27.5 ±0.5 VDC

3.3.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment except as modified herein:

3.3.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve IV of Specification MIL-E-5400.

3.3.9.2 Temperature - The equipment shall operate satisfactorily under the temperature vs. altitude conditions of Specification MIL-E-5400 for continuous operation except that column II of Table I shall not apply. It should be noted that the temperatures specified herein are heat sink temperatures measured at the mounting flange.

3.3.10 Warm-Up Time - The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed 10 seconds under standard conditions and 30 seconds at extreme service conditions.

3.3.11 Input Electrical Power

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.3.11.1 Operating Power - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources having characteristics and limits as defined in MIL-STD-704. The power required shall not exceed the specified amount.

(1) DC Power, 28 V, Category "B", 22 watts

3.3.11.2 Lighting Power - Input power for control panel lighting shall require not more than 0.5 ampere at 5.0 volts.

3.3.11.3 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment.

3.3.12 Cooling - The equipment shall operate satisfactorily under the thermal characteristics described by Curve A of Figure 4 of MIL-E-5400 for Class 2 equipment. No cooling air shall be required.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements of satisfactory performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.4.1 Operation - The augmentor receiver shall receive, amplify and process interrogations from the AN/SPN-42 Radar. Outputs from the receiver shall consist of a trigger, suitable for initiating a response from the aircraft X-Band radar beacon, and a tracking error signal to the X-Band modulator.

The receiver shall include an automatic gain control loop to preserve the amplitude modulation of the received signal over the complete dynamic range of the receiver.

Functions performed by the receiver and by the aircraft X-Band beacon shall be controlled and switchable at the control panel. The control shall contain circuitry for initiating and indicating the results of the Built-In - Test function.

3.5 Detail Requirements

3.5.1 Receiver, Augmentor/R-1623(*)/APN - The receiver shall meet the following requirements:

3.5.1.1 Frequency Coverage - The Receiver shall be capable of receiving and processing interrogation signals in the frequency range of 33.0 GHz to 33.4 GHz.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.5.1.8 Receiver Protection - The receiver shall not be damaged by input signla powers up to +10 dbm with no d-c power applied to the receiver.

3.5.1.9 Contents - The receiver shall include subassemblies that perform the following functions:

1. Receiver Crystal Protector
2. Automatic Gain Control
3. Crystal Video Detector
4. Video Amplifier
5. Delay Stabilizer
6. Power Supply

3.5.1.10 Form Factor - The receiver package including mounting flanges and connectors shall not exceed 5.50 inches wide by 2.75 inches high by 8.00 inches long. All access provisions and connections shall be on the 5.50 inch by 2.75 inch side.

3.4.1.11 Weight - The weight of the receiver shall not exceed 5.0 pounds.

3.5.1.12 Electrical Connections - Connections to external circuits shall be provided as indicated below:

Reference Designation	Receptacle Type	Function
J1	Deutsch DM5605-27-13P-013 (Mates with Deutsch DM9700-27-125-010) or an approved equivalent.	Power, Test and Signal
W1	Mates with UG-599/U Waveguide Flange	Antenna Connection

3.5.2 Modulator, X-Band, MD-817(*)/APN

3.5.2.1 Function - The modulator shall amplitude modulate the output signal from the X-Band beacon AN/APN-154(V). The modulation characteristic shall be representative of the modulation envelope of the interrogating signal from the AN/SPN-42 Radar and shall indicate the angular position of the receiver antenna with respect to the boresight of the conically scanning radar antenna pattern. The modulator shall operate on a pulse to pulse basis so as to be capable of processing interlaced interrogations from both channel of the AN/SPN-42 radar

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.5.1.2 Trigger Sensitivity and Dynamic Range - The receiver shall receive and process at least 99% of the interrogations received at 33.2 GHz that are between - 35 dbm and +40 dbm. Input signal level shall be measured, at the receiver antenna terminal.

3.5.1.3 Bandwidth - The receiver bandwidth shall not be less than 400 MHz as measured at the frequencies corresponding to a triggering sensitivity of - 32 dbm.

3.5.1.4 Input Signal Characteristics - Input signals to the receiver shall have the following characteristics:

Pulse Width	0.2 µsec ± 0.03 usec
PRF	2000 ±100 pps or 4000 ±200 pps
Risetime	0.05 µsec maximum

3.5.1.5 Output Trigger - An output trigger to initiate the X-Band reply pulse shall be provided. The trigger shall have the following characteristics when measured across a 75 ohm resistive load:

Amplitude	9 volts minimum
Risetime	0.1 usec maximum
Pulsewidth	0.5 ±0.2 usec

3.5.1.6 Random Triggers - The receiver shall not generate random triggers (those caused by receiver noise) at a rate greater than 10 pps.

3.5.1.7 Delay - Delay characteristics of the receiver shall be such that the overall delay of the crossband radar augmentor, including 1 µsec dealy in the X-Band radar beacon (paragraph 6.7), shall be as listed below:

Receiver Input Signal Level	Approximate Slant Range from AN/SPN-42	Delay
+39 dbm to +19 dbm	100 ft to 1000 ft	1.500 ±0.010 µsec
+19 dbm to -6 dbm	1000 ft to 24000 ft	1.500 ±0.022 µsec
-6 dbm to -35 dbm	24000 ft to max.	1.5000 ±0.072 µsec

Delay is defined as the time spacing between interrogation signal leading edge at a point 3 db down from maximum to the X-Band reply pulse leading edge at a point 3 db down from maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

without introducing crosstalk between the channels. The modulator shall also provide a detected r-f output pulse representative of the X-Band reply for use in the BIT and delay stabilization circuits of the receiver.

3.5.2.2 Frequency Coverage - The modulator shall be capable of operating over the frequency range of 8800 MHz to 9600 MHz.

3.5.2.3 Input Signal Characteristics - The modulator shall be capable of operating with an r-f input signal having the following chacteristics:

1. Peak Power 600 watts maximum

2. Pulsewidth up to 1.0 μsec

3. PRF 2000 ± 100 or 4000 ±200

3.5.2.4 Impedance - The modulator shall operate into a 50 ohm coaxial transmission line.

3.5.2.5 Insertion Loss - Modulator insertion loss, with no drive applied, shall not exceed 2.0 db.

3.5.2.6 Modulation Characteristics - The modulator shall be capable of 25 percent modulation of the AN/APN-154(V) X-Band signal without clipping.

3.5.2.7 Form Factor - The modulator package, including connectors shall not exceed 0.8 inches by 3.0 inches by 3.0 inches.

3.5.2.8 Weight - Modulator weight shall not exceed 0.5 pound.

3.5.2.9 Electrical Connection - Connections to the modulator shall be provided as indicated below:

Reference Designation	Connector	Function
P1	M39012/32-0001 or equivalent (TNC)	r-f input
J1	GRFF 4060-0005 or equivalent (TNC)	r-f input
P2	Ommispectra Co OSM or equivalent	Error Signal Input
P3	Ommispectra OSM 215 or equivalent	Detector Output

3.5.3 Antenna, Ka-Band AS-2406(*)/APR

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3.5.3.1 Function - The antenna shall serve the function of receiving the AN/SPN-42 interrogations.

3.5.3.2 Pattern - Minimum vertical 3 db pattern coverage shall be 20°. Minimum horizontal 3 db pattern coverage shall be 30°.

3.5.3.2 Polarization - The antenna shall be linearly polarized.

3.5.3.4 Form Factor - The antenna shall be a flared waveguide type with a configuration suitable for flush mounting in the aircraft skin. Maximum overall dimensions shall be 1.88 inches wide by 1.1 inches high by 4.10 inches long.

3.5.3.5 Weight - Antenna weight shall not exceed 0.5 pound

3.5.3.6 Connections - The antenna output terminal shall mate with a standard UG-599/U flange.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Test

(4) Life Tests

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor or an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representa-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

tive may be designated to witness the tests when so desired. Contractors not having adequate facilities to conduct all required test shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The Contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with procedures of Specification MIL-T-5422.

4.2.3 Preproduction (First Article) Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Fabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance, and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accesory material and data necessary to test the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be refurbished by the contractor. After refurbishing, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitive results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Reliability Assurance Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturers Run-in Tests

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-in Test (MRIT) - Prior to submission of the equipment for the all equipment tests described in 4.4.2.2.1, the equipment shall be subjected to pre-conditioning operation in accordance with the environmental conditions of test level III of MIL-STD-781 (less vibrations) for a period of not less than seventy-five (75) hours which shall include a minimum of twelve (2) temperature cycles. An additional ten (10) to twenty-four (24) hours of MRIT shall be conducted on each equipment under the same environmental conditions including vibration. Failure reporting as required by MIL-STD-781 shall commence only with the beginning of the additional 10 to 24 hour test period portion of MRIT. Considerations will be made for relaxation of the requirement if by mutual agreement between the procuring agency and the manufacture, it is determined that the equipment will be assembled with "High Reliability" parts and/or sub-assemblies will be subjected to individual tests adequate to support the ultimate equipment test conditions specified above and in 4.4.2.2.1. However, in no case shall the MRIT be less than ten (10) hours on each equipment with vibration and temperature cycling.

4.4.2 Reliability Assurance Tests - Reliability Assurance Tests shall be conducted using MIL-STD-781. Tests are required by both the Qualification Phase and the Production Acceptance Phase shall be conducted.

4.4.2.1 Qualification Phase - Prior to the acceptance of equipments under the contract or order, three (3) equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan III shall be used.

4.4.2.2 Reliability Production Acceptance Phase Tests - Production equipment shall be tested as outlined in MIL-STD-781 (as modified herein). Test Level F shall be used.

4.4.2.2.1 All Equipment Test - Each equipment produced shall be tested for 75 hours.

To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

III of MIL-STD-781 (Extend the line as necessary to accomodate the data). These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attn: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan III curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.2.3 Test Details - The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.3 Special Test - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the reliability assurance tests.

4.4.3.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

1. On an early equipment after an engineering or material change.

2. Whenever failure reports or other information indicate that additional tests are required (This will be determined by the procuring activity).

4.4.3.2 Scope of Tests - Special tests shall consisit of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.4 Equipment Failure - Should a failure occur during either the reliability assurance or special tests, the following action shall be taken:

1. Determine the cause of failure.

2. Determine if the failure is an isolated case or design effect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

3. Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests.

4. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The Contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests). When reliability tests are conducted, the life test shall be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.5 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the even of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The ciruit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures - The procedures used for conducting preproduction tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modity the tests or require any additional test deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to initial production acceptance shall be reconditioned by the contractor. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor unit it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specifications MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in Specification MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended Use - The equipment described in this specification and the X-Band radar beacon equipment listed in 6.7, when properly interconnected, form a crossband radar augmentor that will operate with either or both channels of the AN/SPN-42 radar. This augmentor extends the tracking capabilities of the radar in relatively heavy rain to greater than four nautical miles. The crossband augmentor receives interrogations from the radar in the frequency range 33.0 to 33.4 GHz and transmits replies at X-Band. These interrogation signals are amplitude modulated due to position errors in the target aircraft flight path. The modulation envelop is detected by the receiver and applied to the modulator attached to the X-Band beacon output. The Ka-band receiver shall be capable of processing interlaced interrogations from both channels of the AN/SPN-42 radar without introducing crosstalk between the channels. Amplitude modulated X-Band replies from the airborne augmentor are received at the AN/SPN-42 radar site by an X-Band receiver where both range and angle tracking information is derived.

6.2 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following

1. Title, number, and date of this specification
2. Selection of applicable levels of packaging and packing (See 5.1)

6.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

1. Contract - The contract shall have precedence over any specification.

2. This Specification - The specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

3. Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.4 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.5 Non-Repairable Subassemblies - As a general rule, non-repairable sub-assemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be hed to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies) the non-repairable subassembly should evidence a Mean-Time-To-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.6 Type Designations - The parentheses () when used in the equipment type designation, will be deleted or replaced by either a number or letter furnished by theprocuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.7. The complete type number shall be used on hameplates, shipping records and instruction books, as applicable.

6.7 Associated Equipment - The augmentor receiver specified herein shall operate in conjunction with the following associated equipment:

6.7.1 Radar Beacon AN/APN-154 (V) - The crossband radar augmentor, as described in 6.1, shall use the following components of the AN/APN-154 (V) as specified in MIL-B-23431A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

Item	Quantity
Receiver-Transmitter, Radar RT-763()/APN-154(V)	1
Duplexer CU-1104/APN-154(V)	1
Cable Assemble, Radio Frequency CG-2574/APN-154(V)	2

To be compatible with the ACLS system requirements, the standard Receiver-Transmitter unit must be modified to provide for external triggering and increased reply rate capability (4000 pps maximum). The augmentor receiver described in this specification requires an X-Band beacon delay of 1.0 $\pm$ 0.1 usec. In addition, beacon delay variation as a function of ambient temperature are assumed to be 1.6 nsec/^{o}C or less.

Project Number 5841-N015

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards
Department (ESSD) Code 93
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:36Z
Check the source to verify that this is the current version before use.